

Summer Internship 25-26
On
Full Stack Developer(Nodejs, Reactjs)

as the partial fulfilment of

Semester - V study

for the degree of

**Bachelor of Science (Information
Technology)**

(Affiliated to Sarvajanik University)

during the academic year 2024 -2025.

Developed By:

Yash Ajaybhai Chandrani

Company Name:

Zero&1 Digital Growth Partner



Sarvajanik University

Shree Ramkrishna Institute of Computer Education and Applied Sciences

Behind P.T Science College, M.T.B. College Campus, Athwalines, Surat-395 001



Shree Ramkrishna Institute of Computer Education and Applied Sciences
Behind P.T. Science College, M.T.B College Campus, Athwalines, Surat -395 001.



CERTIFICATE

DEPARTMENT OF COMPUTER SCIENCE

This is to certify that

Student name

Yash Ajaybhai Chandrani

Enrollment no

SR24BSIT028

has / have successfully completed his / her / their Summer Internship work 24-25 entitled.

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Internship Incharge

HOD
Dept. of Computer Science

Date:

Place: Surat

University Examination:

Enrollment No. : SR24BSIT028

Signature of External Examiners:

Personal Details

Name : Yash Ajaybhai Chandrani

Division : B

Enrl No. : SR2BSIT028

Summer Internship Report(Academic Year: 2025–2026)

Company Details

Internship Title : Full Stack Developer.

Company Name : Zero&1 Digital Growth Partner.

Your Role : Trainee – Developer (Intern)

Internship Duration: 28/04/2026 – 28/25/2026

Email : Rushabh3609@gmail.com

Address : Shop **No. 322**, 3rd, Highfield Ascot, VIP Rd, opp. Palm Avenue, Vesu, Surat, Gujarat 395007.

Contact : +91 7600597366

Internship Completion Certificate



Date: May 28, 2026

Dear Yash A. Chandrani,

This is to certify that **Mr. Yash A. Chandrani** has successfully completed his internship as a **Full-Stack Developer** at **Zero & 1**, Vesu, Surat.

The internship commenced on **April 28, 2026**, and concluded on **May 28, 2026**. During this period, he completed the required **180 working hours** assigned by the organization.

During his tenure, Yash carried out the responsibilities and tasks allocated to him in full-stack development. He has completed all the necessary requirements of this learning and development program.

We wish him the best in his future endeavors.

Sincerely,

FOR ZERO & 1



Rushabh Jain
Proprietor & Director

ZERO & 1
DIGITAL GROWTH PARTNER

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Overview

This project is a secure, responsive, and database-driven full-stack e-commerce web application (Nexora-E-com GitHub: <https://github.com/Yashh-11/Nexora-E-com>) built using Node.js, Express.js, React.js, and MongoDB. It is designed to manage online shopping operations efficiently from user authentication to product discovery, cart handling, secure checkout, and comprehensive admin inventory control.

Key Features:

- **RESTful API Architecture:** Modular Express.js backend routing handling decoupled JSON requests.
- **Secure Authentication:** Password hashing via bcrypt.js and token-based JWT validation.
- **Real-Time Database Sync:** Optimized MongoDB queries using Mongoose schemas for instant inventory updates.
- **Responsive Design:** Clean, modern UI ensuring seamless cross-device compatibility.

Core Modules & User Flow:

- I. User accesses the platform landing page and initializes authentication.
- II. Browses the product catalog with dynamic filtering and categorization.
- III. Adds items to the cart and proceeds through secure checkout simulation.
- IV. Admins log in to manage inventory and monitor platform orders.

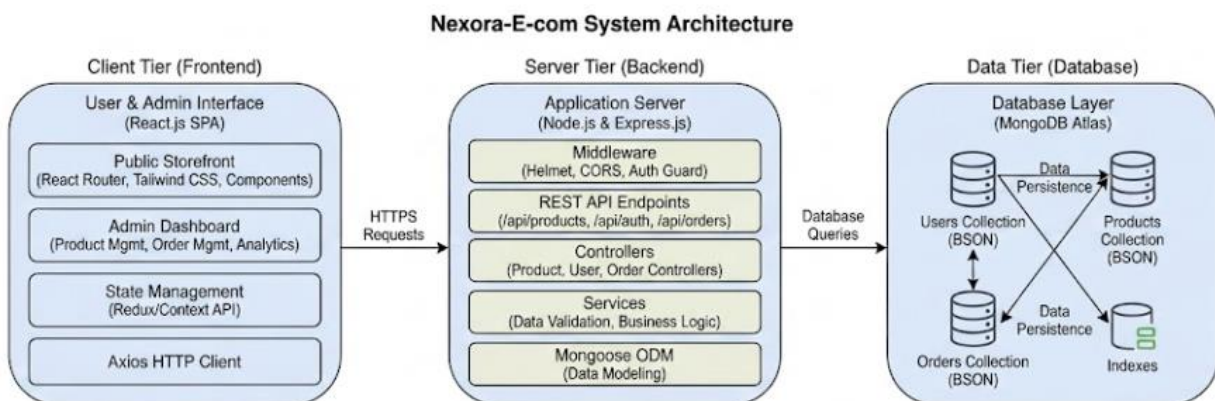
Business Impact :

- **Enhanced User Acquisition & Retention:** The intuitive UI/UX design and responsive multi-device layout reduce bounce rates and create a frictionless shopping experience, directly improving user engagement and customer conversion rates.
- **Streamlined Inventory Operations:** The centralized backend and real-time database synchronization eliminate manual stock tracking errors, ensuring seamless oversight of product quantities and variant statuses.

- **Accelerated Product Discovery:** Advanced server-side search filtering and structured category partitioning drastically reduce search friction, allowing customers to locate desired items in seconds.
- **Robust Data Security & Trust:** Implementing cryptographic password hashing (bcrypt.js) and stateless token authentication (JWT) safeguards user privacy, establishing institutional trust and protecting against unauthorized data breaches.
- **Scalable E-Commerce Architecture:** The decoupled MERN stack (MongoDB, Express, React, Node.js) design ensures that the platform can scale effortlessly to accommodate traffic surges and high-volume transaction loads during promotional events.
- **Optimized Operational Workflow:** The administrative dashboard automates order tracking and catalog updates, reducing administrative overhead and accelerating order fulfillment cycles.

Technologies Used :-

Layer	Technologies Used
Frontend	React.js, React Router, CSS / Bootstrap, Axios, Context API
Backend	Node.js, Express.js, RESTful APIs
Database	MongoDB
Security	JWT, bcrypt.js, Helmet, CORS



2. USER AUTHENTICATION & AUTHORIZATION MODULE

2.1 Module Overview & Security Philosophy

Security is paramount in modern enterprise e-commerce applications. The Authentication and Authorization module of **Nexora-E-com** ensures that customer credentials and administrative privileges remain strictly safeguarded against malicious interception, brute-force attacks, and unauthorized endpoint access. It establishes a trustworthy perimeter for the application.

2.2 Key Features & Security Measures

- **Cryptographic Password Hashing:** Incoming user passwords are automatically salted and hashed using bcrypt.js prior to database persistence, ensuring raw passwords are never exposed.
- **Stateless JWT Sessions:** Eliminates server-side session stores by issuing cryptographically signed JSON Web Tokens upon successful authentication.
- **Role-Based Access Control (RBAC):** Middleware checks user clearances to restrict sensitive API endpoints and admin panels.
- **Input Validation & Sanitization:** Server-side validation rules block malformed requests and neutralize injection vectors.

2.3 Specific Functionality: Credential Verification, Updating, and Session Management

- **Authentication Processing:** Validates incoming username/email and password parameters against MongoDB document records.
- **Token Management:** Issues tokens containing encrypted user IDs and roles, verifying them on subsequent protected requests via Axios interceptors.
- **Profile Updating:** Permits users to update account details, triggering automated re-validation and dynamic token re-issuance.
- **Secure Logout:** Terminates active client-side token states to invalidate user sessions securely.

2.4 User Workflow & Operational Lifecycle

1. The user navigates to the login or registration portal and inputs their credentials.
2. The Express controller verifies data integrity, hashes or compares passwords against stored MongoDB hashes, and generates a signed JWT token.
3. The client application captures the token, stores it securely, and dynamically redirects the user to their designated dashboard or catalog view.

 **Nexora**
Curated commerce

ShopOrdersProfileCart 0UserSignupLogin

CREATE ACCOUNT

Start shopping today

Full name

Email address

Password

Account type

User - browse, buy, and view orders

Create account

Already registered? [Login](#)

 **Nexora**
Curated commerce

ShopOrdersProfileCart 0UserSignupLogin

WELCOME BACK

Login to your account

Email address

Password

Role managed automatically

Your account permissions decide whether you enter the shop or admin studio.

Login

New here? [Create an account](#)

3. REACT FRONTEND & DYNAMIC ROUTING

3.1 Frontend Architecture & Component-Driven Design

The client-side interface of Nexora-E-com is structured using a component-driven architecture in React.js. By breaking down the interface into atomic, reusable components (buttons, cards, navigation bars, modal dialogs, and form inputs), maintainability and testability are dramatically enhanced.

3.2 Key Features & Rendering Strategies

- **Component Modularization:** Clean separation of container components and presentational elements.
- **Client-Side Navigation:** Instant view rendering without full-page browser reloads via react-router-dom.
- **Global State Synchronization:** React Context API manages active user sessions and shopping cart arrays globally.

3.3 Specific Functionality: Dynamic View Switching, Sorting, and Route Guarding

- **Dynamic Routing:** Utilizes route parameters to load specific product detail views, category pages, and user profile panels instantly.
- **Automated Route Guarding:** Inspects global authentication states to restrict unauthorized users from accessing admin routes.
- **State Updates:** Triggers reactive re-renders across components whenever global cart or filter states change.

3.4 User Workflow & Navigation Lifecycle

1. The user clicks a navigation link or triggers an action button within the application interface.
2. React Router updates the browser URL path and mounts the targeted view component.
3. The component lifecycle mounts, dispatches asynchronous Axios requests to the backend, and renders incoming data dynamically.



Nexora
Control studio

[Shop](#)[Orders](#)[Studio](#)[Profile](#)

Cart 1

Admin

Logout



ELECTRONICS

Wireless Headphones

Comfortable over-ear headphones with punchy sound, noise isolation, and fast charging.

Rs. 2,799

Free delivery

22 in stock

Secure checkout

Add to cart

Go to cart



Nexora
Control studio

[Shop](#)[Orders](#)[Studio](#)

Profile

Cart 1

Admin

Logout

Update your account details and review your shopping access.

YOUR ORDERS

7

ACCESS

Read only

ROLE

admin

Profile

Name

Yash

Email

chandraniyash11@gmail.com

Role

admin

Role changes are locked. Admin access is granted only through an invite code during registration.

Save profile

ACCOUNT ACCESS

Admin profile

Use Studio from the navbar when you want inventory and store controls. This page stays focused on your account.

4. PRODUCT CATALOG & ADVANCED SEARCH MODULE

4.1 Module Purpose & Discovery Mechanics

The Product Catalog and Advanced Search module serves as the primary revenue-generating gateway of the platform. It enables users to navigate extensive inventories smoothly through high-speed server-side filtering, text search indexing, and category partitioning.

4.2 Key Features & Indexing Capabilities

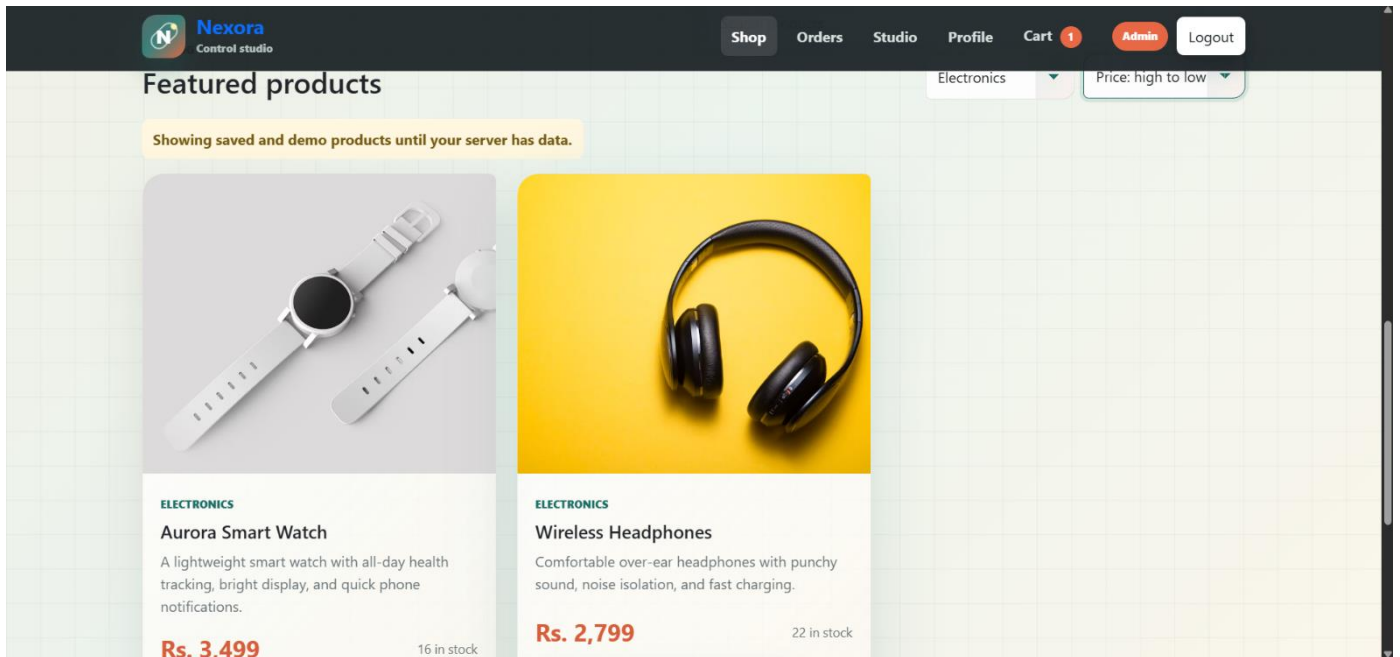
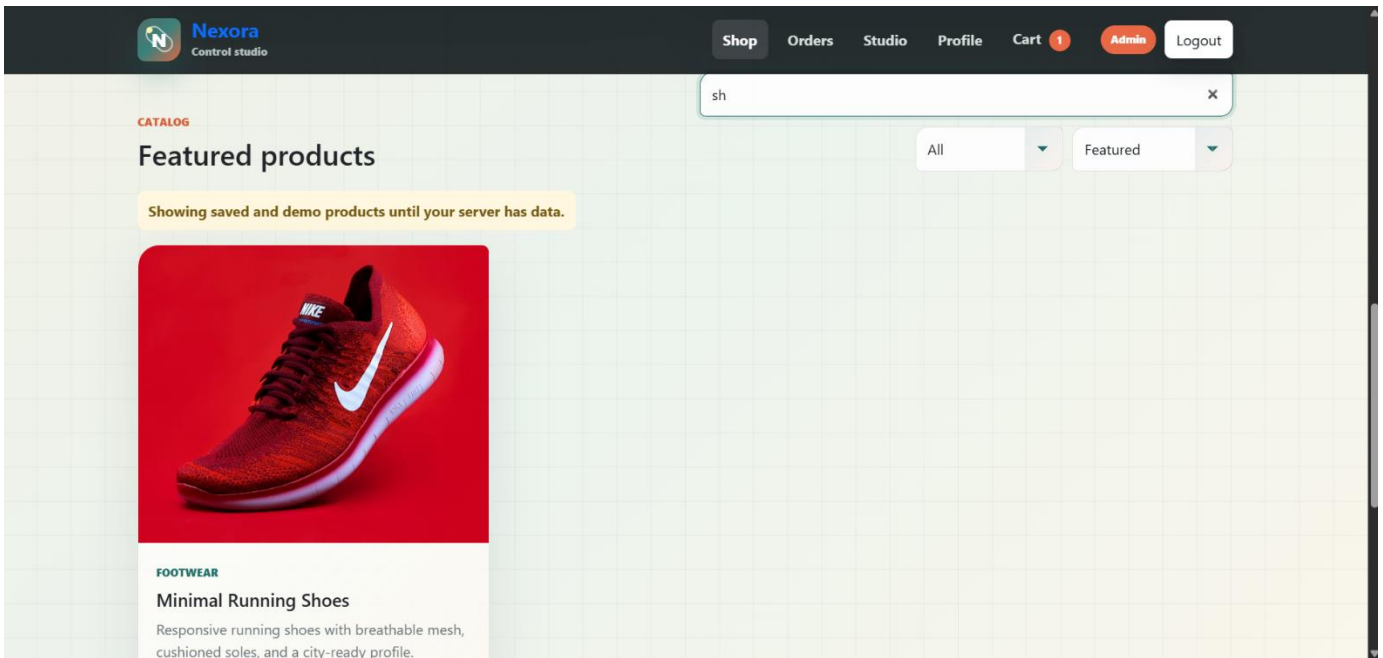
- **Multi-Parameter Filtering:** Filter products simultaneously by category, price range, and keyword.
- **Optimized Text Indexing:** MongoDB text indexes provide rapid lookup performance across product titles and descriptions.
- **Responsive Grid Layout:** Adaptive card displays presenting pricing, ratings, discounts, and action triggers.

4.3 Specific Functionality: Searching, Sorting, and Filtering

- **Advanced Text Search:** Parses user keywords to execute regex and text-match database queries.
- **Multi-Tier Sorting Algorithms:** Sorts inventory dynamically by price (ascending/descending), popularity, or rating scores.
- **Dynamic Pagination:** Breaks large product collections into manageable chunks with total page count metadata.

4.4 User Workflow & Browsing Lifecycle

1. The user inputs keywords into the search bar or selects specific sorting/filtering criteria from dropdowns.
2. Search parameters are transmitted via query strings to the backend search controller.
3. MongoDB executes optimized aggregation queries and returns filtered product chunks for instant client-side rendering



5. SHOPPING CART & SECURE CHECKOUT INTEGRATION

5.1 Transaction Pipeline Overview

The shopping cart and checkout integration manages the critical conversion funnel of the e-commerce platform. It ensures accurate item aggregation, real-time quantity validation, tax/shipping computation, and secure order placement persistence.

5.2 Key Features & Cart Persistence

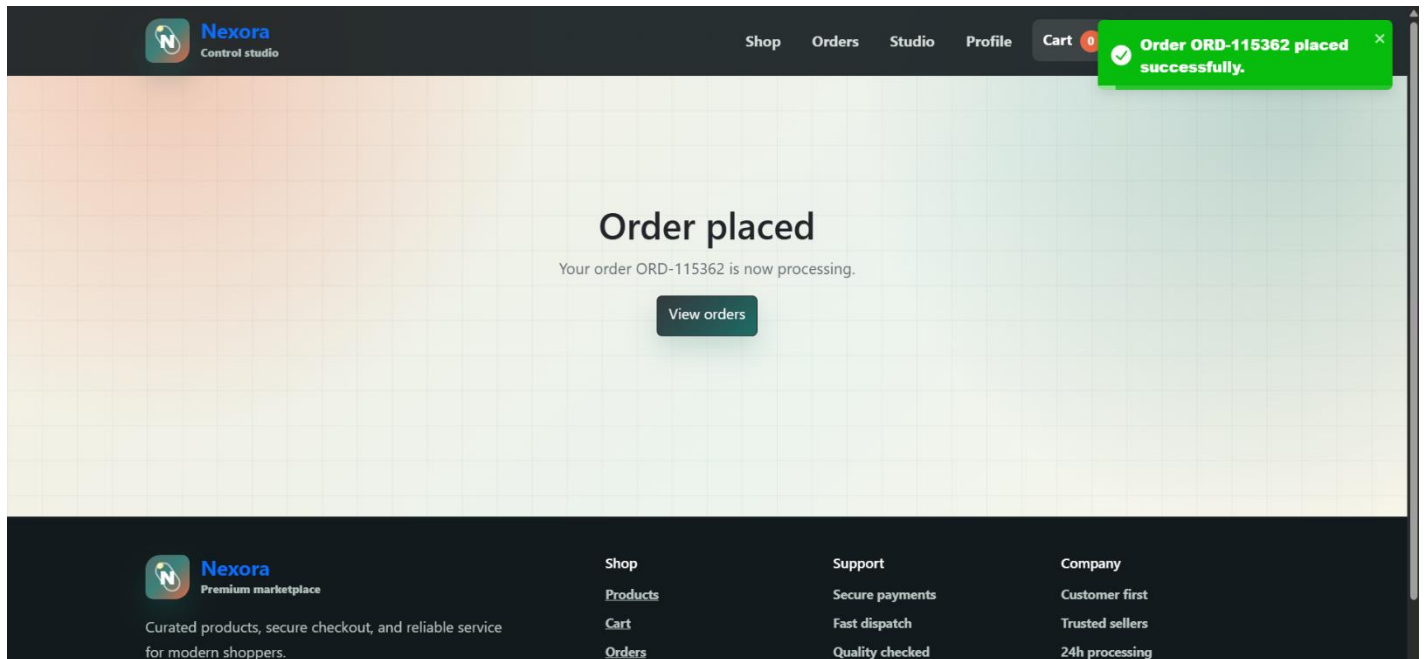
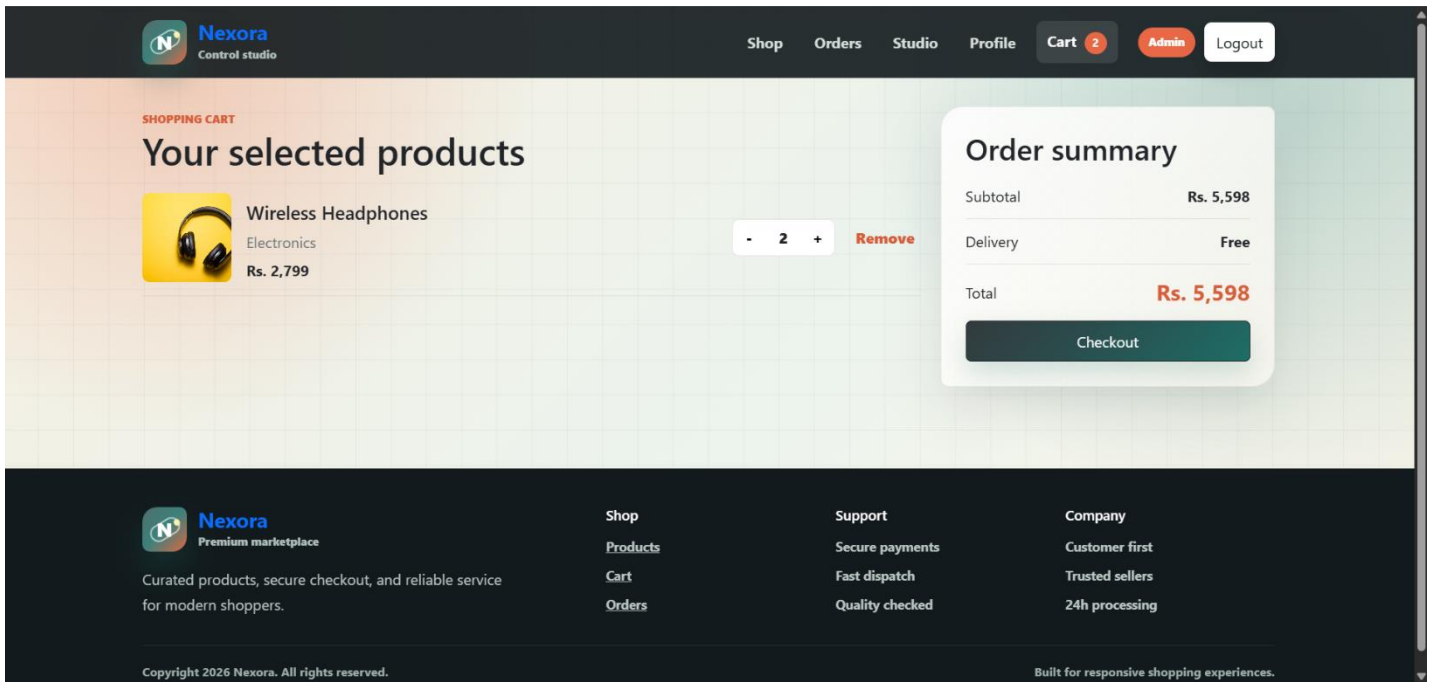
- **Persistent Cloud Cart:** Synchronizes cart contents across user sessions and multiple devices via MongoDB document storage.
- **Real-Time Inventory Validation:** Verifies stock availability against database counts prior to final checkout confirmation.
- **Automated Subtotal Computation:** Computes tax percentages, item discounts, and shipping fees dynamically.

5.3 Specific Functionality: Quantity Updating, Item Deletion, and Checkout Processing

- **Interactive Cart Controls:** Enables users to increment/decrement item quantities or remove unwanted products directly from the cart drawer.
- **Address & Payment Validation:** Validates shipping destination fields and order schemas before committing transactions.
- **Order Persistence:** Records finalized orders into an immutable MongoDB Order collection while clearing active cart states.

5.4 User Workflow & Conversion Lifecycle

1. The user clicks "Add to Cart" on any product listing card.
2. The user reviews selected items in the cart view, modifies quantities, or deletes items.
3. The user proceeds to checkout, inputs shipping details, and submits the order for backend database processing.



6. ADMIN DASHBOARD & INVENTORY MANAGEMENT

6.1 Administrative Control Center Overview

The Admin Dashboard is a heavily protected operational interface granting authorized personnel complete control over platform inventories, product lifecycles, user management, and order fulfillment tracking.

6.2 Key Features & Management Tools

- **Full CRUD Operations:** Seamlessly add, edit, or delete inventory items from the database.
- **Order Fulfillment Tracking:** Real-time oversight of customer order statuses (Pending, Processing, Shipped, Delivered, Cancelled).
- **Sales Analytics Overview:** High-level summaries tracking revenue generation, active user counts, and low-stock alerts.

6.3 Specific Functionality: Inventory Updating, Order Status Modification, and Analytics

- **Inventory Management:** Empowers administrators to update product pricing, descriptions, and stock counts instantly.
- **Order Status Modification:** Updates customer order tracking statuses with immediate database reflection.
- **Analytics Reporting:** Aggregates database collections to display real-time platform metrics and sales performance.

6.4 User Workflow & Administration Lifecycle

1. The administrator authenticates with elevated admin credentials through the secure login portal.
2. The user navigates to the inventory or order management tab within the dashboard interface.
3. The admin executes updates or CRUD actions, which commit instantly to MongoDB and reflect on the customer-facing storefront.

Nexora

Control studio

Products, revenue, inventory, and review store performance

Shop

Orders

Studio

Profile

Cart 0

Admin

Logout

TOTAL ORDERS

8

REVENUE

Rs. 33,390

ROLE

admin

Profile

Name

Yash

Email

chandranish11@gmail.com

Role

admin

Role changes are locked. Admin access is granted only through an invite code during registration.

Save profile

Add product

Product title

Price

Category

Image URL

Description

Nexora

Control studio

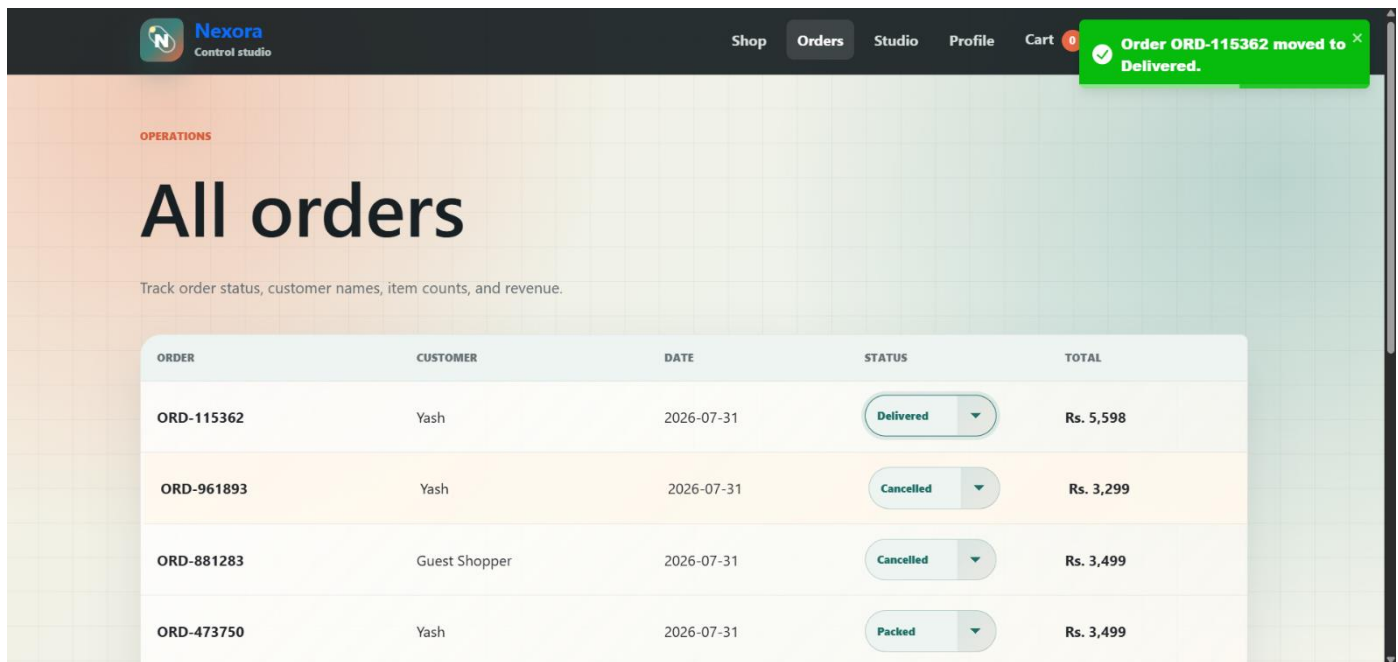
ShopOrdersStudioProfileCart 0AdminLogout

INVENTORY

Product management

PRODUCT	CATEGORY	PRICE	STOCK	ACTION
Aurora Smart Watch	Electronics	Rs. 3,499	18	Delete
Everyday Canvas Tote	Fashion	Rs. 899	34	Delete
Ceramic Desk Lamp	Home	Rs. 1,899	11	Delete
Wireless Headphones	Electronics	Rs. 2,799	22	Delete
Minimal Running Shoes	Footwear	Rs. 3,299	16	Delete
Stoneware Dinner Set	Home	Rs. 2,499	9	Delete

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7. GIT VERSION CONTROL & PROJECT DEPLOYMENT

7.1 Overview & Version Control Strategy

Effective version control and deployment workflows are vital for maintaining code integrity, tracking changes, and delivering applications smoothly to cloud production environments. For **Nexora-E-com**, Git was utilized as the primary version control system to track iterative code additions across frontend and backend modules, while modern cloud pipelines ensured high availability and continuous uptime.

7.2 Key Features & Workflow Integration

- **Branching Strategy:** Maintained isolated feature branches for individual components (e.g., authentication, cart logic, admin dashboard) before merging into production branches via Pull Requests.
- **Ignore Configurations:** Utilized comprehensive .gitignore files to exclude sensitive environment variables (.env), database connection strings, and local build dependencies (node_modules) from public repositories.
- **Repository Management:** Maintained a clean, production-ready GitHub repository ([<https://github.com/Yashh-11/Nexora-E-com>])(<https://github.com/Yashh-11/Nexora-E-com>)).

7.3 Specific Functionality:

Git Operations & Vercel Deployment Pipeline

- **Version Control Commands:** Executed routine version tracking operations including repository initialization (git init), stage tracking (git add), atomic commits (git commit -m), and remote synchronization (git push origin main).
- **MongoDB Atlas Integration:** Configured cloud-hosted MongoDB Atlas database clusters utilizing secure connection URI strings managed via environment variables.
- **Vercel Cloud Deployment:** Configured automated build pipelines on Vercel to detect repository pushes, build React/Node assets, and deploy the full-stack architecture seamlessly.

7.4 User Workflow & Deployment Lifecycle

1. The developer commits tested code modifications locally and pushes updates to the remote GitHub repository (https://github.com/Yashh-11/Nexora-E-com).
2. Vercel automatically triggers a webhook to fetch the latest commit upon every production push.
3. Vercel executes build scripts, injects production environment variables (connecting securely to MongoDB Atlas), and launches the live application.

8. TECHNICAL HIGHLIGHTS

8.1 Module Overview & Architectural Engineering

The technical highlights of Nexora-E-com encompass the advanced engineering paradigms and security practices implemented to ensure high-performance execution across the full stack.

8.2 Key Features & Technical Mechanisms

- **Decoupled MERN Architecture:** Implemented a scalable client-server separation where the Express.js backend exposes clean RESTful JSON APIs consumed asynchronously by a reactive React.js single-page application (SPA).

-
- **MongoDB Atlas Cloud Persistence:** Connected the application to scalable MongoDB Atlas cloud database clusters via Mongoose ODM, ensuring high availability and ACID-compliant document operations.
 - **Cryptographic Security & Hashing:** Integrated bcrypt.js with secure salt rounds within Mongoose pre-save hooks to ensure customer passwords are immutably encrypted prior to database persistence.
 - **Stateless Token Authentication:** Deployed robust JSON Web Token (JWT) session management combined with Axios interceptors, eliminating server-side session overhead and ensuring secure, seamless route authorization.
 - **Vercel Production Deployment:** Optimized project configuration for smooth cloud hosting and serverless/server routing performance on Vercel.
 - **Role-Based Middleware Protection:** Engineered custom Express middleware functions to intercept HTTP requests, verify JWT cryptographic signatures, and enforce strict Role-Based Access Control (RBAC) across sensitive administrator endpoints.

8.3 Specific Functionality: API Optimization & Secure Middleware Interception

- Intercepts incoming HTTP requests to validate bearer tokens before executing controller logic.
- Executes atomic MongoDB Atlas queries to prevent race conditions during inventory modifications.

8.4 User Workflow & Execution Lifecycle

1. An incoming HTTP request hits the deployed server routing tier on Vercel.
2. Custom security and RBAC middleware intercept the request for token verification.
3. Validated requests reach controller functions, executing optimized queries against MongoDB Atlas and returning JSON payloads to the frontend client.

9. CONCLUSION & FUTURE ENHANCEMENTS

9.1 Project Conclusion & Professional Impact

The development and deployment of Nexora-E-com during my summer internship at Zero&1 Digital Growth Partner successfully fulfilled all technical and functional requirements. By integrating Node.js, Express, React, MongoDB Atlas, and Vercel cloud deployment, the application delivers a secure, lightning-fast, and responsive full-stack e-commerce solution. This project bridged academic theory with professional software engineering workflows, cementing my expertise in modern web development.

9.2 Key Features & System Takeaways

- *Robust full-stack architectural decoupling between client presentation and server logic tiers.*
- *Secure stateless authentication utilizing industry-standard cryptographic hashing and token verification.*
- *Cloud-native persistence with MongoDB Atlas and high-performance deployment on Vercel.*

9.3 Specific Functionality & Future Roadmap

Future enhancements include live payment gateway integrations (Stripe/Razorpay), machine learning collaborative filtering recommendation systems, and automated testing pipelines.

9.4 Acknowledgement

I express my profound gratitude to Zero&1 Digital Growth Partner and my mentor Mr. Rushabh Jain for providing me with this exceptional internship opportunity, continuous mentorship, and invaluable technical guidance throughout this project lifecycle.

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ThankYou.